



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL TEST

29-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS, TAAS, TEAS, TIAS, TMAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	C	A	C	C	D	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	A	C	B	B	D	C	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7	34	293	24	60	15	50	13	8	5

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	A	C	A	C	D	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	C	B	A	C	C	D	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	25	5	5	5	3	4	4	18	12	8

PART-3 : MATHEMATICS

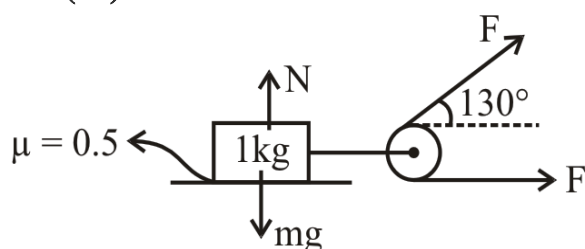
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	D	B	B	B	A	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	A	C	C	D	B	C	B	D	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	220	6	6	2	2	0	4	12	9

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

The block will start slipping if $f_{AB} = f_\ell$.

$$f_\ell = \mu N = \mu [mg - F \sin 30^\circ]$$

$$\text{Also, } F + F \cos 30^\circ = (m + M)a \dots (i) \text{ \& } a = \frac{f_\ell}{M} = \frac{\mu [mg - F \sin 30^\circ]}{M} \dots (ii)$$

From (i) & (ii)

$$F (1 + \cos 30^\circ) = \frac{(m + M) \mu [mg - F \sin 30^\circ]}{M}$$

$$\Rightarrow \frac{20}{11} F \left(1 + \frac{\sqrt{3}}{2} \right) = 10 - \frac{F}{2}$$

$$\Rightarrow F = \frac{10}{3.89} = 2.569 \text{ N}$$

$$\text{Now } F_{\text{applied}} = \frac{F}{2} = 1.2845 \text{ N}$$

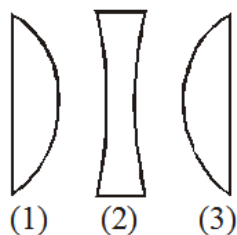
 $\therefore$ Both will move together.

$$a = \frac{1.2845 + 1.2845 \times \frac{\sqrt{3}}{2}}{11} = 0.2179 \text{ m/s}^2$$

$$\therefore f = 2.18 \text{ N}$$

2. Ans (A)

The given system will be equivalent to



$$\text{Given, } \frac{1}{10} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{\infty} + \frac{1}{R}\right)$$

$$\Rightarrow \frac{1}{R} = \frac{1}{5} \Rightarrow R = 5\text{cm}$$

$$\therefore \frac{1}{f_{\text{eq}}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

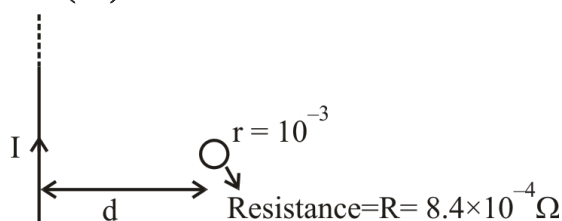
$$\frac{1}{f_2} = \left(\frac{4}{3} - 1\right) \left(\frac{-2}{R}\right) = -\frac{2}{3} \times \frac{1}{R} = -\frac{2}{15}$$

$$\therefore \frac{1}{f_{\text{eq}}} = \frac{1}{10} - \frac{2}{15} + \frac{1}{10} = \frac{3 - 4 + 3}{30} = \frac{1}{15}$$

$\therefore$ Power of the system

$$= \frac{1}{f_{\text{eq}}} = \frac{100}{15} D = 6.67D$$

3. Ans (C)



$$\phi = \frac{\mu_0 I}{2\pi d} \cdot \pi r^2 \Rightarrow \frac{d\phi}{dt} = \frac{\mu_0 r^2}{2d} \frac{dI}{dt}$$

$\therefore$ Current induced

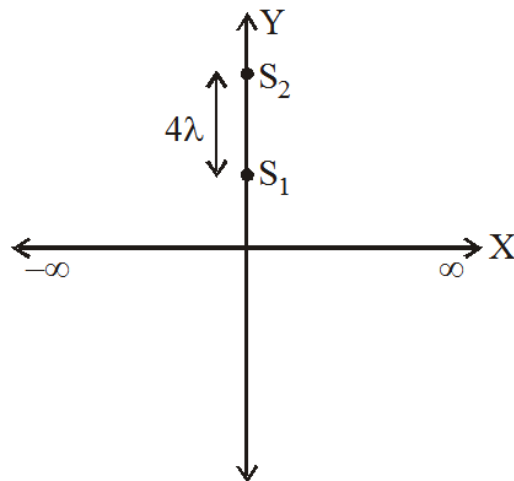
$$= \frac{\mu_0 \times r^2}{2dR} (3.36 \times 2 \times 10^{-12})$$

$\therefore$ Induced current

$$= \frac{4\pi \times 10^{-7} \times 10^{-6}}{2 \times 1 \times 8.4 \times 10^{-4}} \times 3.36 \times 2 \times 10^{-2}$$

$$= 5.024 \times 10^{-11} \text{ A}$$

4. Ans (C)



Maximum path difference = 4λ

Minimum path difference = 0

$\therefore$ No. of minimum = 8

(4 on each side of origin)

5. Ans (A)

Echo is phenomenon of reflection of sound.

6. Ans (C)

Distance upto which signal can be received is

given by $d = \sqrt{2Rh}$

$\therefore$ Area covered = $\pi d^2 = 2\pi Rh$

$$= 2 \times 3.14 \times 6400 \times \frac{1}{10} \text{ km}^2.$$

$\therefore$ Population covered $\approx 4 \times 10^6$.

7. Ans (C)

Since cardboard has to remain stationary

$\therefore f_{\text{net}}$ on cardboard = 0

$$\therefore 3 \times 1 (-\hat{i}) + 2 \times 2 (-\hat{j}) - m\vec{a}_B = 0$$

$$\Rightarrow \vec{a}_B = \frac{-3\hat{i} - 4\hat{j}}{5}$$

$\therefore |\vec{a}_B| = 1 \text{ cm/s}^2$ at 53° South of West.

8. **Ans (D)**

Using work-energy theorem,

$$mg \frac{R}{2} + 0 = \frac{1}{2} I_1 \omega^2$$

$$mg \frac{R}{2} = \frac{1}{2} [I_{\text{ring}} + I_{\text{rod}}]_{\text{Lowest point}} \cdot \omega^2$$

$$mg \frac{R}{2} = \frac{1}{2} \left[(mR^2 + mR^2) + \left(\frac{mR^2}{12} + m \left(R^2 + \frac{R^2}{4} \right) \right) \right] \omega^2$$

$$gm \frac{R}{2} = \frac{1}{2} \left[2mR^2 + \frac{mR^2}{12} + mR^2 + \frac{mR^2}{4} \right] \omega^2$$

$$g = R \left[3 + \frac{1}{12} + \frac{1}{4} \right] \omega^2$$

$$g = R \left(\frac{36 + 1 + 3}{12} \right) \omega^2 = \frac{40}{12} R \omega^2 = \frac{10}{3} \frac{v^2}{R}$$

$$\Rightarrow v = \sqrt{\frac{3gR}{10}}$$

9. **Ans (A)**

$$\Delta m = (1.0087 + 1.0078 - 2.0141) = 0.0024 m_u.$$

$\therefore$ Energy required

$$= 0.0024 \times 931.5 \text{ MeV} = 2.2344 \text{ MeV}.$$

$$\therefore f_r = \frac{E}{h} = \frac{2.2344 \times 10^6 \times 1.6 \times 10^{-19}}{6.625 \times 10^{-34}} \text{ Hz}$$

$$= 0.54 \times 10^{21} \text{ Hz} = 5.4 \times 10^{20} \text{ Hz}$$

10. **Ans (B)**

Total momentum (horizontal) of the water filled inside the tank till time t .

$$P = 2\lambda t v_0$$

Using linear momentum conservation

$$v = \frac{2\lambda t v_0}{M + 2\lambda t}$$

$$v = \frac{2 \times 2 \times 100 \times 5}{100 + 2 \times 2 \times 100} = 4 \text{ m/s}$$

11. **Ans (B)**

From work energy theorem,

$$mgR \sin 37^\circ = \frac{1}{2} \frac{3mR^2}{2} \omega^2 \Rightarrow \omega^2 R = \frac{8g}{5}$$

Also, from torque equation,

$$mgR \cos 37^\circ = \frac{3}{2} mR^2 \alpha$$

$$\therefore \alpha R = \frac{8g}{15}$$

$$\therefore a_B = \sqrt{(\alpha 2R)^2 + (\omega^2 2R)^2} = \frac{8\sqrt{13}}{15} g$$

12. **Ans (B)**

$$\lambda_A = \frac{\ln 2}{0.5} \text{ hr}^{-1} = 2 \ln 2 \text{ hr}^{-1}$$

$$\lambda_B = \frac{\ln 2}{0.25} \text{ hr}^{-1} = 4 \ln 2 \text{ hr}^{-1}$$

$$\therefore \lambda_{\text{net}} = \lambda_A + \lambda_B = 6 \ln 2 \text{ hr}^{-1}$$

Now, at $t = 0$, no. of nuclei = N_0

$$t = 1 \text{ hr, no. of nuclei} = \frac{N_0}{2^2}$$

$$t = 2 \text{ hr, no. of nuclei} = \frac{N_0/2^2}{2^4} = \frac{N_0}{2^6}$$

$$t = 3 \text{ hr, no. of nuclei} = \frac{N_0/2^6}{2^2} = \frac{N_0}{2^8}$$

$$t = 4 \text{ hr, no. of nuclei} = \frac{N_0/2^8}{2^4} = \frac{N_0}{2^{12}}$$

$$t = 5 \text{ hr, no. of nuclei} = \frac{N_0/2^{12}}{2^2} = \frac{N_0}{2^{14}}$$

$$t = 6 \text{ hr, no. of nuclei} = \frac{N_0/2^{14}}{2^4} = \frac{N_0}{2^{18}}$$

$$t = 7 \text{ hr, no. of nuclei} = \frac{N_0/2^{18}}{2^6} = \frac{N_0}{2^{24}}$$

13. **Ans (A)**

We have,

$$E_{\text{eq}} = m\varepsilon$$

$$r_{\text{eq}} = \frac{mr}{n}$$

$$\therefore \text{Current} = \frac{m\varepsilon}{R + \frac{mr}{n}} = \frac{mn\varepsilon}{mr + nR} = \frac{24\varepsilon}{0.5m + 3n}$$

Given $mn = 24$

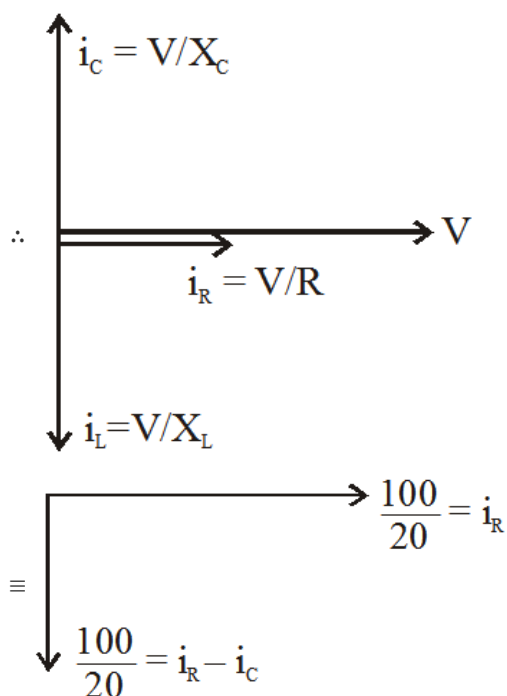
$$\therefore i = \frac{24\varepsilon}{\frac{12}{n} + 3n}$$

For maximum current, $n = 2$

$$\therefore m = 12, n = 2$$

14. Ans (C)

All three are in parallel.



$\therefore$ Net current = $5\sqrt{2}$

$\therefore i_{rms} = 5$

15. Ans (B)

$V_{CE} = 5V$

$\therefore V_{IKW} = 5V \Rightarrow i_C \cdot 1000 = 5$

$\therefore i_C = \frac{5}{1000} A$

$\therefore i_B = \frac{i_C}{\beta} = \frac{5 \times 10^{-3}}{100} = 5 \times 10^{-5}$

Also $V_{RB} = 10 V$

$\therefore 10 = 5 \times 10^{-5} R_B \Rightarrow R_B = 2 \times 10^5 \Omega$

16. Ans (B)

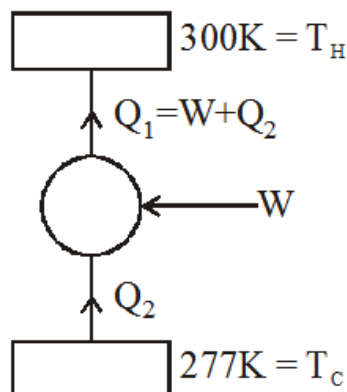
Path diff $\Delta x = \sqrt{40^2 + 9^2} - 40 = 1m$

$\therefore$ For maxima, $\Delta x = n\lambda$.

$\therefore$ For 1st maxima, $n = 1$

$\therefore \Delta x = \lambda \Rightarrow f = \frac{v}{\lambda} = 330Hz$

17. Ans (D)



$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{\frac{T_2}{T_1}}{1 - \frac{T_2}{T_1}} = \frac{T_2}{T_1 - T_2}$$

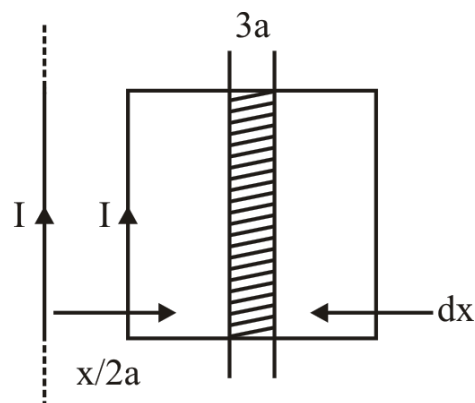
$\therefore \beta = 12$

$\therefore Q_2 = \beta W = 12 J$

$\therefore Q_1 = W + Q_2 = 13 J$

18. Ans (C)

$$w = \int dU = U_f - U_i \left(\because U = \int \mathbf{M} \cdot \mathbf{B} \right)$$



$$U_f = - \int_{2a}^{5a} (I 2a dx) \frac{\mu_0 I}{2\pi x}$$

$$U_i = - \int_{3a}^{4a} (I a dx) \frac{\mu_0 I}{2\pi x}$$

$$w = - \frac{\mu_0 I^2 a}{2\pi} \ln \left(\frac{75}{16} \right)$$

19. Ans (C)

Given circuit is

$$(A + B) \cdot C = Y$$

$\therefore$ For Y to be 1, C must be 1 & either or both of A & B must be 1.

20. Ans (B)

From the given information,

M & L must be ferromagnet while S must be diamagnet.

PART-1 : PHYSICS

SECTION-II

1. Ans (7)



$$F_{\text{air}} \propto v$$

$$F_{\text{air}} = kv$$

$$mu - \int_0^{t_1} kv dt - mgt_1 = 0$$

$$mgt_2 - \int_0^{t_1} kv dt = mv$$

$$mg(t_1 + t_2) = m(v + u)$$

$$T = \left(\frac{v + u}{2} \right) = \frac{70}{10} = 7 \text{ sec.}$$

2. Ans (34)

$$F_{\text{net}} = T - mg$$

$$\therefore dW = \int (T - mg) dy = \Delta KE$$

$$\therefore \frac{1}{2}mv^2 = \int_0^5 T dy - \int_0^5 mg dy$$

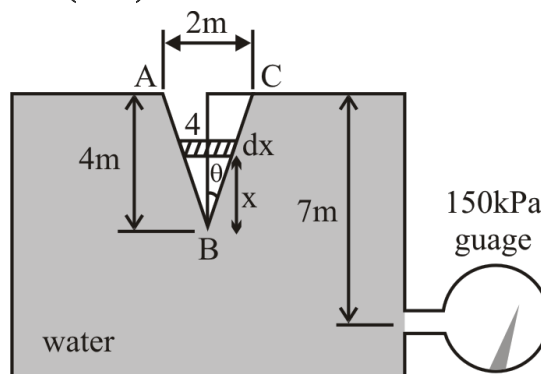
$$T = (6000g - 360g y)$$

$$\therefore \frac{1}{2} \cdot 400v^2 = \int_0^5 (5600g - 360g y) dy$$

$$v^2 = 1175$$

$$v = 34.27$$

3. Ans (293)



$$\tan \theta = \frac{1}{4}$$

$$F_{\text{net}} = \int [150 \times 10^3 - 10^3 \times 10 \times (3 + x)]$$

$$2\pi x \tan \theta \frac{dx}{\cos \theta} \sin \theta$$

$$F_{\text{net}} = \frac{2\pi 10^4}{16} \int_0^4 (15 - 3 + x)x dx$$

$$= \frac{\pi \times 10^4}{8} \left[12 \frac{x^2}{2} - \frac{x^3}{3} \right]$$

$$= \frac{\pi \times 10^4}{8} \left[96 - \frac{64}{3} \right] = 293 \text{ kN}$$

4. Ans (24)

$$\Delta \ell_{Al} = \ell_0 \alpha_{Al} \Delta T$$

$$= 12 \times 12.5 \times 10^{-6} \times 60 = 9 \times 10^{-3} \text{ inch}$$

$$\Delta \ell_{\text{steel}} = 12 \times 6.6 \times 10^{-6} \times 60$$

$$= 4.752 \times 10^{-3} \text{ inch}$$

We have,

$$\tan \theta = \frac{(9 - 4.752) \times 10^{-3}}{1.5} = \frac{\Delta h}{8.5}$$

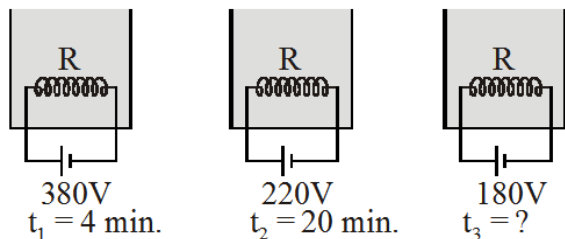
$$\Rightarrow \Delta h = 24.072 \times 10^{-3} \text{ inch}$$

Let reading be h.

$$\therefore h(1 + \alpha_{\text{steel}} \Delta T) = 24.072 \times 10^{-3}$$

$$\Rightarrow h = \frac{24.072 \times 10^{-3}}{1 + \sigma_{\text{Steel}} \Delta T} \approx 24 \times 10^{-3} \text{ inch}$$

5. Ans (60)



$$\left(\frac{V_1^2}{R} - P_{\text{const.}} \right) t_1 = \left(\frac{V_2^2}{R} - P_{\text{const.}} \right) t_2$$

$$= \left(\frac{V_3^2}{R} - P_{\text{const.}} \right) t_3$$

$$\Rightarrow (V_1^2 - \alpha) t_1 = (V_2^2 - \alpha) t_2 = (V_3^2 - \alpha) t_3$$

$$\Rightarrow V_1^2 t_1 - V_2^2 t_2 = \alpha (t_1 - t_2)$$

$$\therefore \alpha = \frac{V_2^2 t_2 - V_1^2 t_1}{t_2 - t_1} = 24400$$

$$\Rightarrow (220^2 - 24400) 20 = (180^2 - 24400) t_3$$

$$\therefore t_3 = 60 \text{ min.}$$

6. Ans (15)

In 0.5g potassium, ^{40}K will be $\left(\frac{0.0117}{100} \times 0.5 \right) \text{g}$

$$\therefore \text{No. of } ^{40}\text{K-nucleus} = \frac{0.5 \times 0.0117}{100 \times 40} \times 6 \times 10^{23}$$

$$= 8.775 \times 10^{17}$$

$$\therefore \text{Activity} = \lambda N = \frac{\ln 2}{t_{\frac{1}{2}}} N = 15.22 \text{ dps}$$

7. Ans (50)

The capacitors $2\mu\text{F}$ and $1\mu\text{F}$ are in parallel combination.

So, let $1\mu\text{F}$ get a charge q_1 and $2\mu\text{F}$ get a charge of q_2 .

$$q_1 : q_2 = C_1 : C_2 \\ = 1 : 2$$

So, given $q_2 = 20 \mu\text{J}$

So, $q_1 = 10 \mu\text{J}$

$$\text{So, energy stored in } C_1 = \frac{10^2}{2 \times 1} = 50 \mu\text{J}$$

$$\text{and energy stored in } C_2 = \frac{20^2}{2 \times 2} = 100 \mu\text{J}$$

So, total energy stored in capacitor

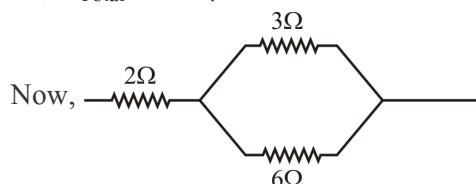
$$= 100 + 50 = 150 \mu\text{J}$$

Now, if we calculate energy dissipated in resistors from the formulation

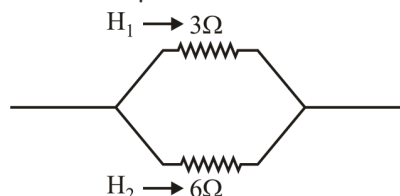
$$H = \int i^2 R \, dt$$

It will be same as E_{stored} in capacitor

So, $H_{\text{Total}} = 150 \mu\text{J}$



Energy dissipated in the branch of 3Ω & 6Ω will be $75 \mu\text{J}$



$$H_1 : H_2 = 6 : 3 = 2 : 1$$

$$\text{So, } H_1 = \frac{2}{3} \times 75 = 50 \mu\text{J}$$

8. Ans (13)

For object to be resolved,

$$\frac{1.22\lambda D}{a} \geq d$$

$$\Rightarrow \frac{1.22 \times 580 \times 10^{-9} \times D}{7 \times 10^{-3}} = 1.3$$

$$\Rightarrow D = \frac{1.3 \times 7 \times 10^{-3}}{1.22 \times 580 \times 10^{-9}} \text{ m} = 0.01286 \times 10^6 \text{ m} \\ = 13 \text{ km}$$

9. Ans (8)

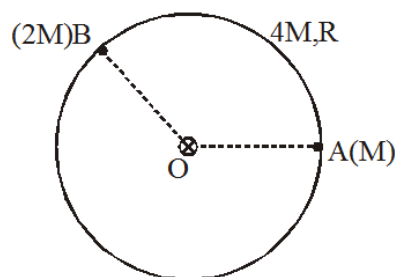
Given,

$$V = \frac{2V_0}{T}t - V_0$$

$$\therefore V_{\text{rms}} = \frac{\int V^2 dt}{\int dt} = \frac{\int_0^T \left(\frac{2V_0}{T}t - V_0 \right)^2 dt}{\int_0^T dt}$$

$$= \frac{\left[\frac{T \left(\frac{2V_0}{T}t - V_0 \right)^2}{3 \times 2V_0} \right]_0^T}{T} = \frac{2V_0^3}{6V_0} = \frac{V_0^2}{3} = 8$$

10. Ans (5)



τ_0 on the system = 0

Let disc rotate by angle θ

Case I for θ to be minimum,

$$2MR^2 \left(\frac{2\pi}{3} - \theta \right) - MR^2 \left(\frac{2\pi}{3} + \theta \right) - \frac{4MR^2}{2} \theta = 0$$

$$\Rightarrow \frac{4\pi}{3} - 2\theta - \frac{2\pi}{3} - \theta - 2\theta = 0 \Rightarrow \theta = \frac{2\pi}{15}$$

Use II : for θ to be maximum,

$$2MR^2 \left(\frac{4\pi}{3} - \theta \right) + MR^2 \left(\frac{2\pi}{3} - \theta \right) - \frac{4MR^2}{2} \theta = 0$$

$$\Rightarrow \frac{8\pi}{3} - 2\theta + \frac{2\pi}{3} - \theta - 2\theta = 0$$

$$\Rightarrow \theta = \frac{10\pi}{15}$$

$$\therefore \frac{\theta_{\text{max}}}{\theta_{\text{min}}} = 5$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A)



no of moles of eletron lost by 1 mole of As_2S_3
 $= 2 \times 2 + 3 \times 8 = 28$

Electron lost by 5 mole = $28 \times 5 = 140$

50 % electron = 70 mole

Now, $\text{A}^{x+} \rightarrow \text{A}^{x-}$

Cange in oxidation number = $2x$

Now $35 \times 2x = 70$

$x = 1$

2. Ans (C)

Half-life for trimerisation process

$$t_{\frac{1}{2}} = \frac{0.693}{K} = \frac{0.693}{6.93 \times 10^{-2}} = 10 \text{ min}$$

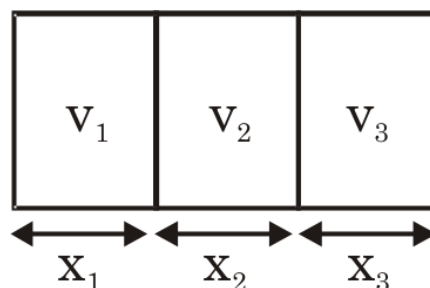
No of moles of B undergoes trimerisation

$$= 2 \times \frac{1}{2} = 1 \text{ mole}$$

Now, $P = P_A^0 \times x_A$

$$= 60 \times \frac{5}{6} = 50 \text{ mm Hg}$$

3. Ans (D)



Initially the two piston are 10 cm apart

When temperature becomes 400 K, 500 K, 600

K, then $v_1 : v_2 : v_3 = 4 : 5 : 6$ is

$$A \times x_1 : A \times x_2 : A \times x_3 = 4 : 5 : 6$$

(A = cross sectional area)

$$x_1 : x_2 : x_3 = 4 : 5 : 6 \text{ and}$$

$$x_1 + x_2 + x_3 = 30 \text{ cm}$$

$$\Rightarrow x_1 = 8 \text{ cm}, x_2 = 10 \text{ cm}, x_3 = 12 \text{ cm}$$

The difference of separation between two pistons = $10 - 10 = 0$

4. **Ans (A)**

$$K_p = (P_{H_2O})^2$$

$$\Rightarrow 4 \times 10^{-2} = (P_{H_2O})^2$$

$$\Rightarrow P_{H_2O} = 2 \times 10^{-2} \text{ atm}$$

$$= 760 \times 2 \times 10^{-2}$$

$$= 15.20 \text{ atm}$$

City A having relative humidity = 60%

$$\text{aqueous tension} = 20 \times \frac{60}{100} = 12 \text{ mm Hg}$$

$$\text{City B the aqueous tension} = 20 \times \frac{80}{100}$$

$$= 16 \text{ mm Hg}$$

5. **Ans (C)**

Fact based

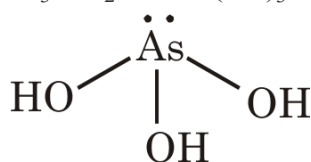
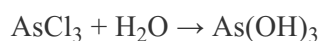
6. **Ans (A)**



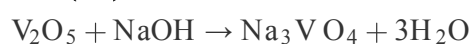
7. **Ans (C)**

SO_3^{-2} , SCN^- & CN^- exhibit linkage isomers

8. **Ans (D)**



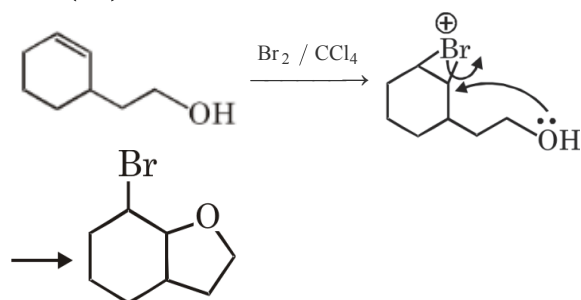
9. **Ans (A)**



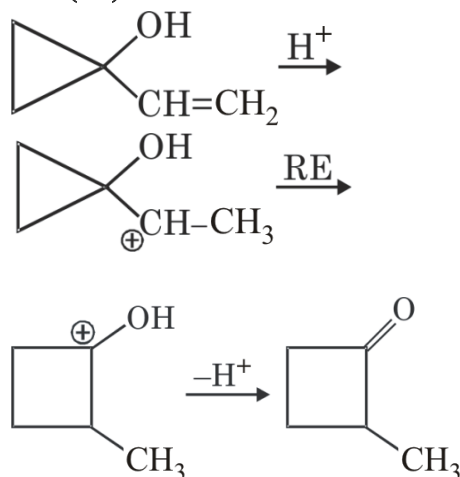
11. **Ans (B)**

ClO_2 is about 30 times stronger bleaching agent, diamagnetic
it disproportionate in basic medium.

12. **Ans (C)**



13. **Ans (C)**



19. **Ans (C)**

Number of atoms of A in unit cell

$$= 4 - \frac{1}{8} \times 4 = \frac{7}{2}$$

Number of atoms of B in unit cell

$$= \frac{1}{4} \times 12 = 3$$

Number of atoms of C in unit cell = 4

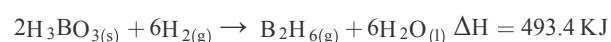
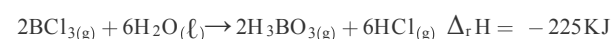
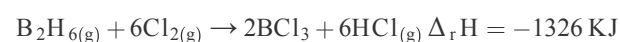
$$x = 7, y = 6, z = 8$$

$$x + y + z = 21$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (25)**



$$\text{Molar heat of sublimation} = \frac{50}{2} = 25 \text{ KJ/mole}$$

3. **Ans (5)**

Let initial concentration of acid is c and 1st step degree of ionisation is α .

Then $[\text{H}^+]$ concentration

$$= c\alpha + \frac{c\alpha}{2} + \frac{c\alpha}{4} + \dots + \frac{c\alpha}{2^{n-1}}$$

$$= c\alpha \left(1 + \frac{1}{2} + \dots + \frac{1}{2^{n-1}} \right)$$

$$= c\alpha \left(\frac{1 - \left(\frac{1}{2}\right)^n}{\frac{1}{2}} \right)$$

$$(\text{A}^{n-}) = \frac{c\alpha}{2^{n-1}}, \text{ then } n = 5$$

5. Ans (3)

For $H_xFe(CO)_4$, $18e^-$ rule for Fe = $8 - (-x) + 4 \times 2$
= 18

$$\Rightarrow x + 8 = 10$$

$$\Rightarrow x = 2$$

Similarly, Re belongs to 7th group.

So, $18e^-$ rule for Re = $7 - (-y) + 10 = 18$

$$\Rightarrow 7 + y = 8$$

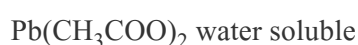
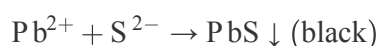
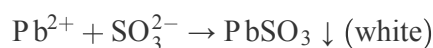
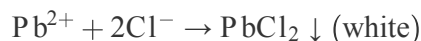
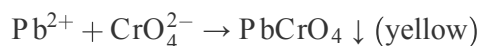
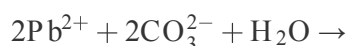
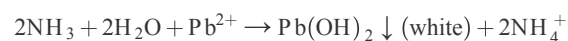
$$\Rightarrow y = 1$$

$$\Rightarrow x + y = 3$$

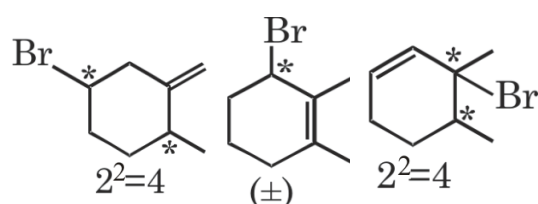
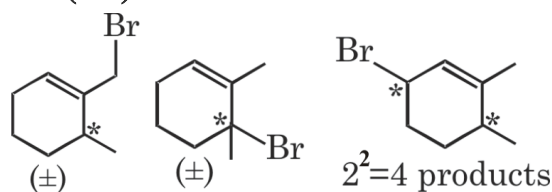
6. Ans (4)

(a), (c), (f), (i) are correct.

7. Ans (4)



8. Ans (18)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$\tan \theta + \sqrt{3} \tan \theta \cdot \tan 2\theta = \sqrt{3} - \tan 2\theta$$

$$\Rightarrow \frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \cdot \tan 2\theta} = \sqrt{3}$$

$$\Rightarrow \tan 3\theta = \tan \frac{\pi}{3}$$

$$\Rightarrow 3\theta = n\pi + \frac{\pi}{3}$$

$$\Rightarrow \theta = \frac{n\pi}{3} + \frac{\pi}{9}$$

No. of solution is $\left[-\pi, \frac{\pi}{2}\right]$ is 5.

2. Ans (B)

$$x^2 - \left(5 + 3\sqrt{\log_3 5} - 5\sqrt{\log_5 3}\right)x + 3 \left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0$$

$$\therefore 3\sqrt{\log_3 5} = 3\sqrt{\log_3 5} \cdot \sqrt{\log_3 5} \cdot \sqrt{\log_3 5} = 3^{\log_3 5} \cdot \sqrt{\log_3 5}$$

$$= (3^{\log_3 5})^{\sqrt{\log_3 5}} = 5\sqrt{\log_5 3}$$

$$\text{and } 3^{\sqrt[3]{\log_3 5}} = 3^{\log_3 5 \cdot \sqrt[3]{(\log_3 5)^2}} = (3^{\log_3 5})^{(\log_3 5)^{2/3}}$$

$$= 5^{(\log_5 3)^{2/3}}$$

$\therefore$ Equation reduces to $x^2 - 5x - 3 = 0$ $\begin{matrix} \alpha \\ \beta \end{matrix}$

$$\text{Let } x = \frac{1}{\alpha}$$

$$\alpha = \frac{1}{x}$$

$$\therefore \frac{1}{x^2} - \frac{5}{x} - 3 = 0$$

$$\Rightarrow 3x^2 + 5x - 1 = 0$$

3. Ans (B)

Let $s \equiv \sin t$

$c \equiv \cos t$

Let orthocentre be $H(h, k)$

$\therefore$ Equilateral triangle

$\Rightarrow$ Orthocentre = Centroid

$$\therefore h = \frac{s + c + a}{3}$$

$$\Rightarrow 3h - a = s + c$$

$$\text{and } k = \frac{-c + s + b}{3}$$

$$\Rightarrow 3k - b = s - c$$

$$\therefore (3h - a)^2 + (3k - b)^2 = (s + c)^2 + (s - c)^2$$

$$\begin{aligned} & A(s, -c) \\ & B(c, s) \quad C(a, b) \\ & = 2(s^2 + c^2) \\ & = 2 \end{aligned}$$

$$\Rightarrow \left(h - \frac{a}{3}\right)^2 + \left(k - \frac{b}{3}\right)^2 = \frac{2}{9}$$

$$\therefore \text{locus is } \left(x - \frac{a}{3}\right)^2 + \left(y - \frac{b}{3}\right)^2 = \frac{2}{9}$$

$$\text{Centre } \left(\frac{a}{3}, \frac{b}{3}\right)$$

$$\text{Given, centre } \left(\frac{2}{3}, \frac{1}{3}\right)$$

$$\therefore a = 2 \quad b = 1$$

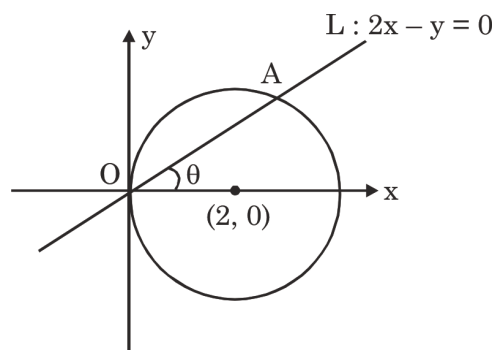
$$\text{Hence } (a^2 + b^2) = 4 + 1$$

$$= 5$$

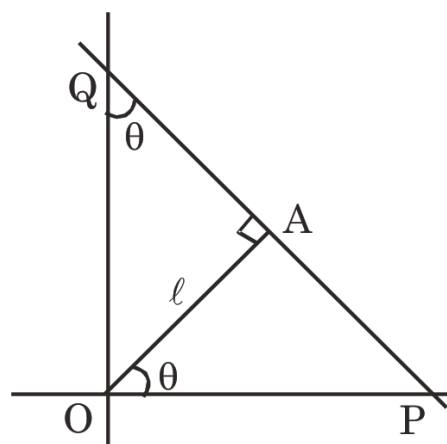
4. Ans (D)

$$C_1 : (x - 2)^2 + y^2 = 2^2$$

$$\tan \theta = \frac{y}{x} = 2$$



C_2 is a circle with OA as diameter.



$\therefore$ tangent at A is perpendicular to OA

$$\text{let } OA = \ell$$

$$\therefore QA = \ell \cot \theta$$

$$PA = \ell \tan \theta$$

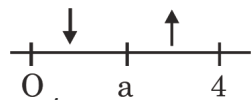
Hence,

$$\frac{AP}{PQ} = \frac{\ell \tan \theta}{\ell \tan \theta + \ell \cot \theta} = \frac{2}{2 + \frac{1}{2}} = \frac{4}{5}$$

5. Ans (B)

$$f(x) = 2x^2 - \ln x$$

$$f'(x) = 4x - \frac{1}{x}$$



$$\therefore f'(a) = 0$$

$$\Rightarrow 4a - \frac{1}{a} = 0$$

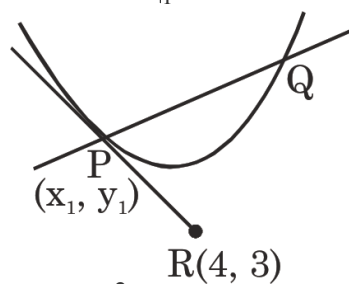
$$\Rightarrow a = \frac{1}{2}$$

$$\therefore R(8a, 8a - 1) = (4, 3)$$

$$\therefore \text{Parabola is } y^2 = 2x$$

Now, any point on parabola be $P(x_1, y_1)$

$$m_{PR} = \left. \frac{dy}{dx} \right|_P$$



$$\Rightarrow \frac{y_1 - 3}{x_1 - 4} = \frac{1}{y_1}$$

$$\Rightarrow y_1^2 - 3y_1 = x_1 - 4$$

$$\Rightarrow y_1^2 - 3y_1 = \frac{y_1^2}{2} - 4$$

$$\Rightarrow y_1^2 - 6y_1 + 8 = 0$$

$$\Rightarrow y_1 = 2, 4$$

$$\therefore P(8, 4) \text{ or } P(2, 2) \text{ rejected.}$$

$$\text{Let } P(8, 4) = \left(\frac{1}{2}t_1^2, t_1 \right)$$

$$\therefore t_1 = 4$$

$\therefore$ Normal at $P(t_1)$ meets the parabola again at $Q(t_2)$.

$$\therefore t_2 = -t_1 - \frac{2}{t_1}$$

$$= -4 - \frac{1}{2} = -\frac{9}{2}$$

$$\therefore \text{Slope of tangent at } Q(t_2) = \frac{1}{t_2} = -\frac{2}{9}$$

6. Ans (B)

$$\sin^{-1} x = k\alpha \text{ and } \cos^{-1} x = k\beta$$

$$\therefore \frac{\pi}{2} = k(\alpha + \beta)$$

$$k = \frac{\pi}{2(\alpha + \beta)}$$

$$\sin\left(\frac{\pi\alpha}{\alpha + \beta}\right)$$

$$= \sin(2\sin^{-1} x)$$

$$= 2 \sin(\sin^{-1} x) \cdot \cos(\sin^{-1} x)$$

$$= 2x \cdot \sqrt{1 - x^2}$$

7. Ans (B)

$$\lim_{x \rightarrow \infty} (8x^3 + mx^2)^{1/3} - nx$$

$$= \lim_{x \rightarrow \infty} (2x) \left(1 + \frac{m}{8x}\right)^{1/3} - nx$$

$$= \lim_{x \rightarrow \infty} (2x) \left(1 + \frac{m}{24x}\right) - nx$$

$$= \lim_{x \rightarrow \infty} (2 - n)x + \frac{m}{12}$$

For limit to exist $2 - n = 0$

$$\Rightarrow n = 2 \text{ and } \frac{m}{12} = 1$$

$$\Rightarrow m = 12$$

$$\therefore \frac{m}{n} = \frac{12}{2} = 6$$

8. Ans (A)

$$f(g(x)) = x, \text{ so } f'(g(x)) g'(x) = 1.$$

Putting $x = a$, we get

$$f'(g(a)) g'(a) = 1 \Rightarrow f'(b) = \frac{1}{2}.$$

9. Ans (C)

Let the circumcentre of the triangle be the origin.

$\Rightarrow$ Orthocentre is $\vec{a} + \vec{b} + \vec{c}$ and the centroid is

$$\frac{\vec{a} + \vec{b} + \vec{c}}{3}.$$

$$\Rightarrow F = \frac{2}{3}(\vec{a} + \vec{b} + \vec{c})$$

$$\Rightarrow (\vec{a} - F)^2 + (\vec{b} - F)^2 + (\vec{c} - F)^2$$

$$= \frac{1}{9} [(\vec{a} - 2(\vec{b} + \vec{c}))^2 + (\vec{b} - 2(\vec{a} + \vec{c}))^2 + (\vec{c} - 2(\vec{a} + \vec{b}))^2]$$

$$= \frac{1}{9}(27) = 3$$

10. **Ans (B)**

$$\text{Required probability} = \frac{{}^7C_2}{{}^7C_2 + {}^7C_1 \cdot {}^5C_1} = \frac{3}{8}$$

11. **Ans (D)**

Consider unit circle $|z| = 1$ with a regular polygon of 9 vertices inscribed in it with one of

the vertices as 1. 'v' and 'w' are any two distinct

$$\text{vertices such that } \left| \frac{v+w}{2} \right| \geq \frac{1}{2}.$$

$$\text{This implies that } \left| \arg\left(\frac{v}{w}\right) \right| \leq \frac{2\pi}{3} \Rightarrow \left| \arg\left(\frac{v}{w}\right) \right| \leq \frac{6\pi}{9}.$$

Thus, the total number of pairs v and w can be

$$\frac{(6+5+4+3+2+1)}{{}^9C_2} = \frac{21}{36} = \frac{7}{12}$$

12. **Ans (A)**

$$\sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} = \sqrt{\frac{124}{48}}$$

13. **Ans (C)**

Eliminate n form given equations

$$al^2 + bm^2 + c\left(\frac{-(pl + qm)}{r}\right)^2 = 0$$

$$\Rightarrow (ar^2 + cp^2)\left(\frac{\ell}{m}\right)^2 + 2cpq\left(\frac{\ell}{m}\right) + br^2 + cq^2 = 0$$

The lines are parallels then DCs are equal then

roots are equal

$$B^2 - 4AC = 0$$

$$\Rightarrow [2cpq]^2 = 4(ar^2 + cp^2)(br^2 + cq^2)$$

$$\Rightarrow abr^2 + acq^2 + bcp^2 = 0$$

$$\Rightarrow \frac{p^2}{a} + \frac{q^2}{b} + \frac{r^2}{c} = 0$$

$$\Rightarrow \sum \frac{p^2}{a} = 0$$

14. **Ans (C)**

$$\frac{1}{x^2} + \frac{1}{f^2(x)} = 1$$

$$\Rightarrow f(x) = \frac{x}{\sqrt{x^2 - 1}}$$

$$\Rightarrow f(f(x)) = x$$

$$\Rightarrow f(f(f(x))) = f(x) = \frac{x}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \underbrace{fff \dots - f(x)}_{10 \text{ times}} = x$$

$$\text{So } \int_2^4 \underbrace{fff \dots f(x)}_{10 \text{ times}} dx = \left[\frac{x^2}{2} \right]_2^4 = \frac{16-4}{2} = 6$$

15. **Ans (D)**

$$E_n : \frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1 \quad a_n > b_n$$

$$b_n^2 = a_n^2(1 - e_n^2)$$

$$\text{Given } b_n = b_{n-1} \text{ \& } a_{n-1} = a_n e_n$$

$$\text{So } b_{n-1}^2 = a_n^2(1 - e_n^2)$$

$$\text{For } E_{n-1} : a_{n-1}^2 = b_{n-1}^2(1 - e_{n-1}^2)$$

$$a_n^2 e_n^2 = a_n^2(1 - e_n^2)(1 - e_{n-1}^2)$$

$$e_n^2 = (1 - e_n^2)(1 - e_{n-1}^2)$$

$$\Rightarrow e_n = \sqrt{\frac{1 - e_{n-1}^2}{2 - e_{n-1}^2}}$$

$$\Rightarrow e_2 = \sqrt{\frac{1 - e_1^2}{2 - e_1^2}} = \sqrt{\frac{1 - \frac{3}{4}}{2 - \frac{3}{4}}} = \frac{1}{\sqrt{5}}$$

$$\Rightarrow e_3 = \frac{2}{3} \quad e_4 = \sqrt{\frac{5}{14}}$$

16. Ans (B)

$$\lim_{x \rightarrow 0} \left(1 + \frac{f(x)}{x^3} \right)^{1/x} \text{ exists so } \lim_{x \rightarrow 0} \frac{f(x)}{x^3} = 0$$

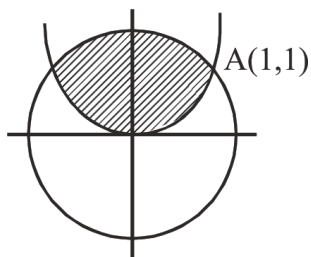
$$\text{So } f(x) = a_4 x^4 + a_5 x^5 + \dots + a_n x^n$$

$$\text{For least degree } f(x) = a_4 x^4$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{f(x)}{x^3} \right)^{1/x} = e \Rightarrow a_4 = 1 \Rightarrow f(x) = x^4$$

$$\text{So we have } y = x^4 \quad x^2 + y^2 = 2$$

$$\text{Area} = 2 \int_0^1 (\sqrt{2-x^2} - x^4) dx = \frac{\pi}{2} + \frac{3}{5}$$



17. Ans (C)

$$\alpha + \gamma = \frac{4}{t_1}, \quad \alpha\gamma = \frac{1}{t_1}, \quad \beta + \delta = \frac{6}{t_2}, \quad \beta\delta = \frac{1}{t_2}$$

$\alpha, \beta, \gamma, \delta$ are in H.P.

$$\Rightarrow \beta = \frac{2\alpha\gamma}{\alpha + \gamma} = \frac{2/t_1}{4/t_1} = \frac{1}{2}$$

$$\gamma = \frac{2\beta\delta}{\beta + \delta} = \frac{2/t_2}{6/t_2} = \frac{1}{3}$$

$$\gamma = \frac{1}{3} \text{ root of } t_1 x^2 - 4x + 1 = 0 \Rightarrow t_1 = 3$$

$$\beta = \frac{1}{2} \text{ root of } t_2 x^2 - 6x + 1 = 0 \Rightarrow t_2 = 8$$

Normal to $xy = c^2$ at $(ct, \frac{c}{t})$ is

$$xt^3 - yt - ct^4 + c = 0$$

$$\text{Slope} = t^2$$

So product of slopes

$$= t_1^2 \cdot t_2^2 = (9)(64) = 576$$

18. Ans (B)

$$[(p \wedge q) \vee (\sim p \wedge q) \vee (r \wedge \sim q)] \vee [p \wedge (\sim p \vee \sim q) \wedge q]$$

$$= [(p \vee \sim p) \wedge q] \vee [r \wedge \sim q] \vee [(p \wedge \sim p) \vee (p \wedge \sim q) \wedge q]$$

$$= [T \wedge q] \vee [r \wedge \sim q] \vee [(F \vee (p \wedge \sim q)) \wedge q]$$

$$= (q \vee r) \wedge (q \vee \sim q) \vee [p \wedge (\sim q \wedge q)]$$

$$= (q \vee r) \wedge T \vee [P \wedge F]$$

$$= ((q \vee r) \wedge T) \vee F = q \vee r$$

19. Ans (D)

$$\frac{PB}{AP} = \frac{2}{1}$$

Let A (h, 0) and B (0, K)

Using section formula

$$(1) \quad \frac{2h+0}{3} = x \Rightarrow h = \frac{3}{2}x$$

$$(2) \quad \frac{K+0}{3} = y \Rightarrow K = 3y$$

$$\Rightarrow A \left(\frac{3}{2}x, 0 \right), \quad B (0, 3y)$$

$$\text{Slope of tangent} \Rightarrow \frac{dy}{dx} = \frac{3y}{-\frac{3}{2}x}$$

$$x \frac{dy}{dx} + 2y = 0$$

Solving different equation

$$\frac{dy}{y} + \frac{2dx}{x} = 0$$

$$f(x) = \frac{1}{x^2}$$

Normal (1, 1) is $2y = x + 1$

20. Ans (B)

Let $a-d, a, a+d$ are 3 roots of equation

$$\text{Sum of root} = (a-d) + a + (a+d) = 12$$

$$3a = 12$$

$$a = 4$$

$$\text{Product of roots} : a(a-d)(a+d) = 28$$

$$a(a^2 - d^2) = 28$$

$$a = 4 \text{ put in equation } \therefore k = 39$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5)

$$\text{Given : } 2\cos x + 2\cos 2x + \sin 2x + 2\cos x \sin 2x$$

$$= 2\sin x$$

$$= 2\cos x + 2\cos 2x + 2\sin x \cos x + (\sin 3x - \sin x)$$

$$= 0$$

$$= 2\cos x + 2\cos 2x + 2\sin x \cos x + 2\cos 2x \sin x$$

$$= 0$$

$$= 2\cos x (1 + \sin x) + 2\cos 2x (1 + \sin x) = 0$$

$$(1 + \sin x) (2\cos x + 2\cos 2x) = 0$$

$$2(1 + \sin x) (\cos x + \cos 2x) = 0$$

$$2(1 + \sin x) 2\cos\left(\frac{3x}{2}\right) \cdot \cos\left(\frac{x}{2}\right) = 0$$

$$\Rightarrow \sin x = -1 \Rightarrow x = 2n\pi + \frac{3\pi}{2}$$

$$\boxed{x = \frac{-\pi}{2}} \text{ are solution in } [-\pi, \pi]$$

$$\cos \frac{3x}{2} = 0 \Rightarrow \frac{3x}{2} = (2n+1) \frac{\pi}{2}$$

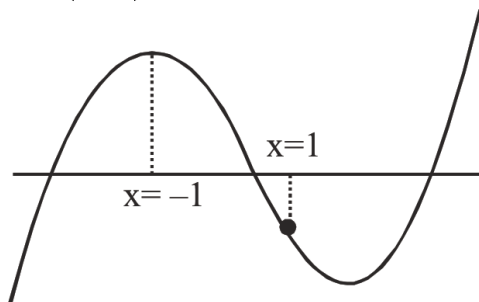
$$x = (2n+1) \frac{\pi}{3} \quad \frac{\pi}{3}, \frac{-\pi}{3}, \pi, -\pi$$

$$\cos \frac{x}{2} = 0, \Rightarrow x = (2n+1) \pi = \pi, -\pi$$

$$\text{All solutions} = x = \left\{ -\pi, \frac{-\pi}{2}, \frac{-\pi}{3}, \frac{\pi}{3}, \pi \right\}$$

5 solutions.

2. Ans (220)



$$p(x) \rightarrow \text{cubic}$$

$$p'(x) \rightarrow \text{quadratic}$$

$$p'(x) \text{ min at } x = 1$$

$$\Rightarrow p''(x) = 0 \text{ at } x = 1$$

$$p''(x) = \lambda(x - 1)$$

$$p'(x) = \lambda\left(\frac{x^2}{2} - x\right) + c$$

$$p'(-1) = 0 \text{ as } p(x) \text{ take maximum at } x = -1$$

$$c = -\frac{3}{2}\lambda$$

$$p'(x) = \lambda\left(\frac{x^2}{2} - x - \frac{3}{2}\right)$$

$$p(x) = \lambda\left(\frac{x^3}{6} - \frac{x^2}{2} - \frac{3}{2}x\right) + k$$

$$p(1) = \lambda\left(\frac{1}{6} - \frac{1}{2} - \frac{3}{2}\right) + k = -6$$

$$\lambda = 6, \quad k = 5$$

$$p(x) = 6\left(\frac{x^3}{6} - \frac{x^2}{2} - \frac{3}{2}x\right) + 5$$

$$p(x) = x^3 - 3x^2 - 3x + 5$$

$$p(3) = 27 - 27 - 27 + 5$$

$$p(3) = -22$$

$$10p(3) = -220$$

3. Ans (6)

Given system of eq. have non-zero solution.

$$D = 0$$

$$D = \begin{vmatrix} t-1 & 3t+1 & 2t \\ t-1 & 4t-2 & t+3 \\ 2 & 3t+1 & 3t-3 \end{vmatrix} = 0$$

$$= D = 6t(t-3)^2$$

$$t = 0 \text{ and } t = 3$$

Smallest value of $t = 0$

$$-x + y = 0 \quad \dots (i)$$

$$-x - 2y + 3z = 0 \quad \dots (ii)$$

$$2x + y - 3z = 0 \quad \dots (iii)$$

From eq. (1) : $-x + y = 0$

$$x = y$$

Put these $x = y$ in eq. (ii)

$$-x + 2y + 3z = 0$$

$$-3x + 3z = 0$$

$$x = z$$

we can say

$$x = y = z$$

Put in eq. (iii) and check

$$3x - 3z = 0 \text{ satisfies}$$

$$x : y : z = 1 : 1 : 1$$

$$a = 1 = b = c$$

$$a + b + c = 3$$

Sum of all diff. value of $t = 0 + 3 = 3$

$$\text{Ans. } 3 + 3 = 6$$

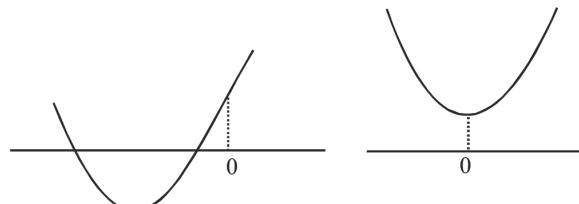
4. Ans (6)

$$f(x) = 9x^3 + 9(m-1)x^2 + (m+5)x + 7$$

$$f'(x) = 27x^2 + 18(m-1)x + 3(m+5)$$

$$f'(x) = 3[9x^2 + 6(m-1)x + (m+5)]$$

$$f'(x) \geq 0 \text{ for } x \geq 0$$



$$(1) \left(\frac{-b}{2a} < 0 \right)$$

$$\frac{-6(m-1)}{9} \leq 0$$

$$\frac{6(m-1)}{9} \geq 0$$

$$m \geq 1 \quad \dots (i)$$

$$(2) f(0) \geq 0$$

$$m + 5 \geq 0$$

$$m \geq -5 \quad \dots (ii)$$

From (i) and (ii),

$$\Rightarrow m \in [1, \infty)$$

$$k = 1, 6k^2 = 6$$

$$\text{Ans. } 6$$

5. Ans (2)

$$\text{Let } A - B = C \Rightarrow \text{tr}(C) = 0$$

Characteristic equation of matrix C is

$$\lambda^2 - \text{tr}(C) \cdot \lambda + |C| = 0$$

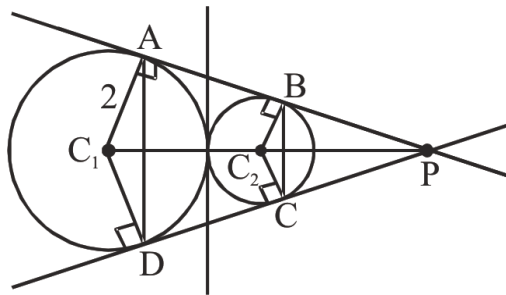
$$\therefore C^2 - (\text{tr}(C))C + |C|I = 0$$

$$\Rightarrow \text{tr}(C^2) - (\text{tr}(C))^2 + |C| \times 2 = 0$$

$$\Rightarrow \frac{\text{tr}(C^2)}{|C|} = -2$$

$$\Rightarrow \frac{\text{tr}((A-B)^2)}{\det(A-B)} = -2$$

6. Ans (2)



$$\Rightarrow \text{Length of intercept} = \frac{AD + BC}{2}$$

$$C_1P = 6; \quad C_2P = 3$$

$$\Rightarrow AD = 2 \times 2 \times \frac{2\sqrt{2}}{3} = \frac{8\sqrt{2}}{3}$$

$$BC = 1 \times 2 \times \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{3}$$

Now intercept made by radical axis

$$= \frac{AD + BC}{2} = \frac{6\sqrt{2}}{3} = 2\sqrt{2}$$

7. Ans (0)

Let x be the highest power of 3 in $n!$.

$$\therefore x = \sum_{k=1}^{\infty} \left[\frac{n}{3^k} \right] < \sum_{k=1}^{\infty} \frac{n}{3^k}$$

$$\therefore x < \frac{n/3}{1 - 1/3} \Rightarrow x < \frac{n}{2}$$

Now $x = 117$

$$\therefore n > 2 \times 117 \Rightarrow n > 234$$

For $n = 237$,

$$\left[\frac{237}{3} \right] + \left[\frac{237}{9} \right] + \left[\frac{237}{27} \right] + \left[\frac{237}{81} \right] = 115$$

$\therefore$ When $n = 240$ highest power of 3 will be

116 for $n = 241$, $n = 242$ this will remain same

but when $n = 243$, highest power will be 121.

$\therefore$ There does not exist any n .

8. Ans (4)

$$N = 68600 = 2^3 \times 5^2 \times 7^3$$

$$= 1 \times (2^3 \cdot 5^2) \times 7^3 = 1 \times (2^3 7^3) \times 5^2$$

$$= 1 \times (5^2 \times 7^3) \times 2^3 = 2^3 \times 5^2 \times 7^3$$

9. Ans (12)

Given functional equation is

$$(f(x) - x)^2 + (f(y) - y)^2 = (f(x + y) - (x + y))^2$$

$$\text{Let } g(x) = (f(x) - x)^2$$

$$\text{Now } g(x) + g(y) = g(x + y)$$

$$\therefore g(x) = kx = (f(x) - x)^2$$

$$f(1) = 2 \Rightarrow k = 1$$

$$\therefore (f(9) - 9)^2 = 9 \Rightarrow f(9) = 12$$

10. Ans (9)

$$f(x) = \begin{cases} \left[\frac{x-3}{x-2} \right], & x < 1 \\ k, & x = 1 \\ x^2 + ax + b, & x > 1 \end{cases}$$

at $x = 1$, LHL = 1

$$f(1) = k$$

$$\text{RHL} = 1 + a + b$$

$$\therefore k = 1 \text{ and } a + b + 1 = 1$$

at $x = 1$ LHD = 0

$$\text{RHD} = 2 + a$$

$$\therefore a = -2 \Rightarrow b = 2$$